GTFN: GCN and Transformer Fusion Network With Spatial-Spectral Features for Hyperspectral Image Classification

Aitao Yang, Min Li, Yao Ding, Danfeng Hong, *Senior Member, IEEE*, Yilong Lv, and Yujie He

***Abstract*— Transformer has been widely used in classification tasks for hyperspectral images (HSIs) in recent years. Because it can mine spectral sequence information to establish long-range dependence, its classification performance can be comparable with the convolutional neural network (CNN). However, both CNN and Transformer focus excessively on spatial or spectral domain features, resulting in an insufficient combination of spatial-spectral domain information from HSI for modeling. To solve this problem, we propose a new end-to-end graph convolutional network (GCN) and Transformer fusion network (GTFN) with the spatial-spectral feature extraction in this article, which combines the strengths of GCN and Transformer in both spatial and spectral domain feature extraction, taking full advantage of the contextual information of classified pixels while establishing remote dependencies in the spectral domain compared with previous approaches. In addition, GTFN uses Follow Patch as an input to the GCN and effectively solves the problem of high model complexity while mining the relationship between pixels. It is worth noting that the spectral attention module is introduced in the process of GCN feature extraction, focusing on the contribution of different spectral bands to the classification. More importantly, to overcome the problem that Transformer is too scattered in the frequency domain feature extraction, a neighborhood convolution module is designed to fuse the local spectral domain features. On Indian Pines, Salinas, and Pavia University datasets, the overall accuracies (OAs) of our GTFN are 94.00%, 96.81%, and 95.14%, respectively. The core code of GTFN is released at https://github.com/1useryang/GTFN.**

***Index Terms*— Graph convolution neural network (GCN), hyperspectral image (HSI) classification, transformer.**

I. Introduction

HYPERSPECTRAL images have hundreds of near-continuous spectra with a wealth of spectral–spatial information. The extremely high spectral resolution makes it possible for hyperspectral image (HSI) to distinguish feature information of different objects at a fine-grained level. Therefore, HSI classification is widely used in remote sensing, such as fine agriculture [16], mineral analysis [18], urban planning [19], and military target monitoring [20].

In the HSI classification, each image pixel has hundreds of spectral bands. This high-dimensional information helps to improve the classification on the one hand and increases the dimensionality of feature extraction on the other hand. Also, spatial relationships of different features in HSI also provide critical information for classification, so efficiently extracting hyperspectral spatial-spectral information has essential research significance in HSI classification.

In the early period of HSI classification, plenty of machine learning-based approaches were generated, including random forests [21], K-nearest neighbor methods (KNN) [23], Markov random fields [22], and support vector machine radial basis function (SVM-RBF) [24]. KNN and SVM perform well as classical spectral domain classifiers when dealing with high-dimensional data and have excellent classification capabilities. However, the traditional algorithms rely on manual design, require extensive experience and some expertise, and lack powerful data fitting capabilities to extract deep semantic features. With the development of deep learning (DL), DL approaches have been widely applied to HSI classification to overcome the shortcomings of traditional methods, such as autoencoders (AEs) [25], convolutional neural networks (CNNs) [1], graph convolutional networks (GCNs) [28], long- and short-time memory (LSTM) [29], recurrent neural networks (RNNs) [26], and Transformer [30].

The DL approaches can automatically extract high-dimensional features and achieve end-to-end classification compared with traditional methods. CNN has become the most widely used model for extracting features from HSI. To extract spectral domain features of HSI images, a 1-D CNN [1] was proposed, where a neural network with five convolutional layers was designed to classify HSI in the spectral domain. However, the 1-D CNN ignored the abundant spatial information in HSI images, so a 3-D CNN was proposed [31] to combine spectral–spatial information better. To better handle the 3-D information of HSIs, a multiscale 3-D deep CNN (M3D-DCNN) is proposed in [32], which can better learn 1-D spectral features and 2-D spatial features. With the deepening of CNN's network layers, networks are prone to problems such as gradient disappearance and overfitting. To eliminate these adverse effects, a deep feature fusion network (DFFN) was proposed by Song et al. [33], and the DFFN improves the

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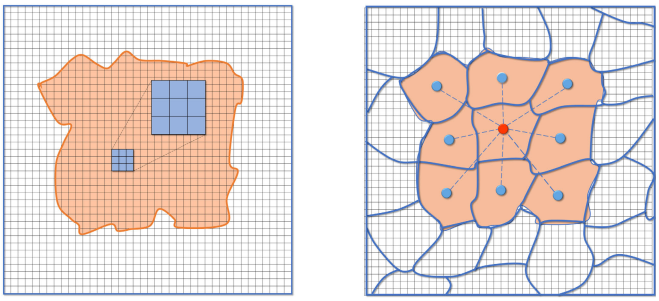
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(a) (b)

Fig. 1. Illustration of CNN and GCN. The blue squares in (a) CNN convolution region, where the CNN extracts local spatial information and generates pixel-level features. (b) GCN operation after superpixel segmentation of the image. The dots denote the superpixel centers, the connecting lines denote the neighborhood relationships, and the GCN can convolve irregular regions and establish pixel remote dependencies. The orange area indicates classified objects.

accuracy of HSI classification by using strong complementary information between different convolutional layers. To better focus on spectral domain features, Zhu et al. [34] proposed a residual spectral–spatial attention network (RSSAN), suppressing worthless bands, focusing on essential features, and designing a spatial-attention module to achieve an adaptive selection of spatial information. To avoid feature extractors only focusing on spatial or spectral information, Li et al. [14] proposed a double-brand dual-attention mechanism network (DBDA), which jointly captured spatial-spectral features of HSIs by designing the spatial attention module and the channel attention module.

The CNN-based classification model automatically captures the features of HSI in the spatial-spectral domain, which improves the classification performance. However, it is difficult to establish long-term dependencies due to the local connectivity of the CNN, and it is difficult to extract information from spectral band sequences explicitly. Moreover, the fixed-size convolution kernel is not flexible enough to handle arbitrarily shaped regions. In addition, expanding the receptive field by stacking convolutional layers will increase the computational overhead, and the number of convolutional layers will limit the size of the receptive field.

Compared with CNN networks, graph convolution neural network (GCN) can aggregate features of neighboring nodes in non-Euclidean space to achieve convolutional operations, thus establishing remote dependencies of different nodes in space, which significantly improves the ability of pixel points to combine spatial contextual information, as shown in Fig. 1. Therefore, the utilization of GCNs into HSI classification has become a research hotspot. In [10], a graph CNN in the null spectral domain (S2GCN) was proposed, approximating the convolution using the neighboring nodes of the graph, which significantly improved the classification performance but was computationally intensive because each pixel is considered a node. To reduce the amount of calculation, Hong et al. [11] quantitatively and qualitatively analyzed the role of CNNs and GCNs in image classification and proposed miniGCN, which enables the training of large-scale GCNs in a small-scale manner. To exploit the spatial and spectral correlation property in multiscale space, a multiscale dynamic GCN (MDGCN) was proposed in [12] to construct multiple input graphs with different neighborhood scales.

In recent years, the CNN and GCN fusion-based methods of joint extraction of information have been proposed to overcome the shortcomings of small samples of CNN and the huge computational cost faced by GCN. CEGCN [43] fully leveraged the advantages of CNN and GCN, achieving joint mining of pixel-level and superpixel-level spatial-spectral features and achieving complementary spectral feature extraction. Dong et al. [15] established a superpixel-based graph attention networks (GAT) and proposed a weighted feature fusion of CNN and graph attention network (WFCG) for HSI classification using an attention mechanism CNN through weighted fusion. Zhou et al. [27] proposed a novel multiple-feature fusion model termed attention multihop graph and multiscale convolutional fusion network (AMGCFN), which aggregated context information from the network through multihop GCN, captured the characteristics of different size receptive fields through multiscale CNN, and designed an attention fusion module to realize the adaptation of the two networks.

Although GCN improves the classification performance of HSI, there are still certain shortcomings: constructing the adjacency matrix incurs a high computational cost and reduces the classification efficiency; the adjacency matrix only establishes the relationship between pixels in space, ignoring the correlation of the spectrum. To overcome the shortcomings caused by CNNs, Vaswani et al. [5] proposed a new architecture called Transformer for processing sequence information. Transformer uses a self-attention (SA) mechanism to overcome the effects of distance and establish sequence remote dependencies, thus focusing on globally relevant information of the image and less on irrelevant details. To introduce the Transformer architecture into computer vision, Dosovitskiy et al. [6] proposed the Vision Transformer (ViT) to input images as patches into the Transformer, achieving better classification accuracy compared with previous CNN models. The Transformer learns data features through the multihead self-attention (MHSA) to learn the data features, which could establish the global relationship of the image compared with CNN. As a result, the Transformer has superior capabilities in capturing long-term image dependencies. In recent years, due to the superiority of Transformer's SA mechanism, Transformer has also been implemented in HSI classification tasks (see [7], [35], [36]). To address the problems of weak flexibility and poor generalization of traditional DL approaches in HSI classification problems, He et al. [7] proposed the HSI-BERT, which supported flexible and dynamic inputs and captured global dependencies between global pixels without considering the spatial distance of pixels. In [9], a spatial-spectral feature tokenization transformation (SSFTT) was proposed to capture spatial-spectral features, and the feature transformation was performed by a Gaussian weighted feature tokenizer, which better represented the high-level semantic features, and the algorithm efficiency was greatly improved. To address the problem of overdispersion of features extracted by MHSA in Transformer, a group-aware hierarchical transformer (GAHT) was proposed in [30], which introduced a grouped pixel

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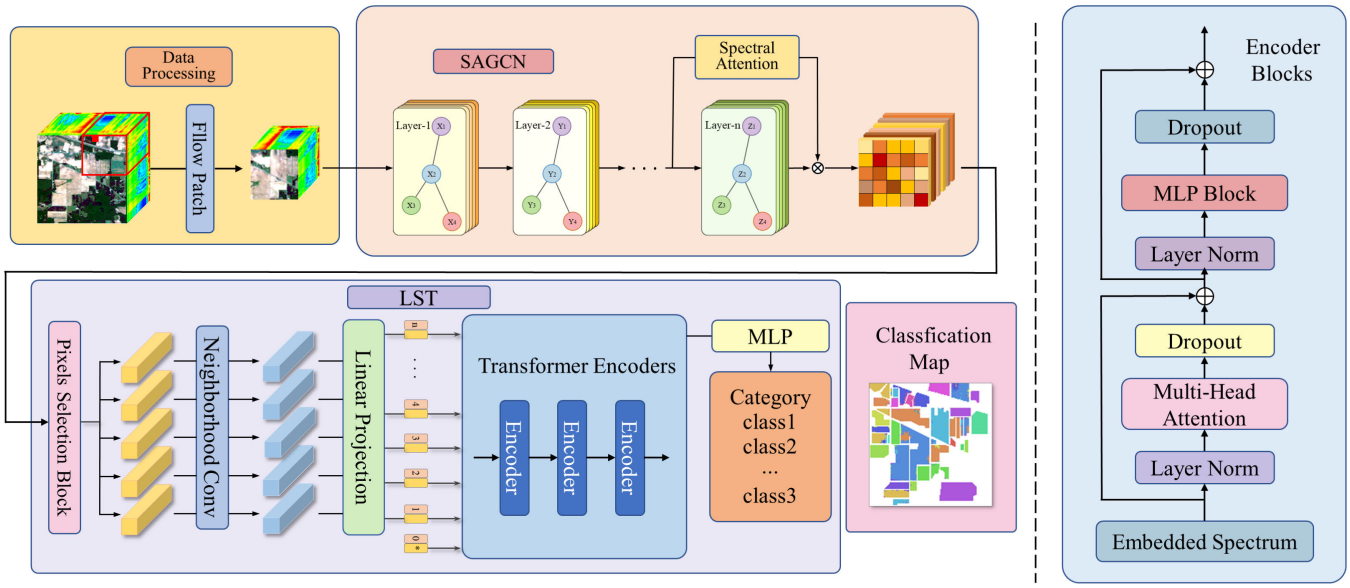


Fig. 2. Overview illustration of the proposed GTFN. The network's core consists of three parts: the data processing, the SAGCN subnetwork, and the LST subnetwork. The blue area on the right is the structure of the encoder in the LST.

embedding (GPE) module that allowed MHSA to better focus on local spatial-spectral information.

Transformer can model hyperspectral spectral domain information well, so it has excellent capabilities in HSI classification. However, Transformer also has the following drawbacks: first, compared with its application in visible images, HSI images have a multitude of bands, and the abundant spectral domain information could lead to the extraction of features that are too scattered, ignoring the contribution of local neighboring spectral bands to object classification [30]. In addition, in extracting spatial semantic features, Transformer cannot model the relationship between adjacent pixels, and the global SA mechanism of MHSA is only limited to spectral domain feature extraction. Moreover, to exploit the image's spatial information, the spatial-spectral domain fusion Transformer often uses image patches as the token, which can lead to cross-category boundary pixel misclassification as the input is not a single pixel, and the undifferentiated input neighborhood information can also lead to high computational complexity.

To solve the above problems and to better exploit the spatial-spectral domain features of HSI while exploiting the ability of GCN to model different nodes and the Transformer's ability to establish long-term dependencies on spectral domain sequences, this article fully combines the advantages of both models and proposes a new spatial-spectral domain classification network, GTFN, as shown in Fig. 2. Specifically, we design the Follow Patch module to reduce the computational complexity of GCN better. After that, to improve the ability of the spatial-spectral domain information of GCN, we introduce the spectral attention module and propose the spectral attention GCN (SAGCN). In addition, we design the pixel selection module to choose the essential output features of GCN and simplify the input of Transformer. Finally, in extracting the features of Transformer, we creatively propose the neighborhood convolution to take advantage of the global attention mechanism of Transformer while focusing better on local spectral information. The main innovative contributions are summarized as follows.

1) We design a novel end-to-end spatial-spectral depth learning classification framework GTFN, which combines the advantages of GCN and Transformer with spatial-spectral features for HSI classification.
2) We introduce a spectral attention module into GCN to automatically learn the importance of each spectral segment, and enhance the spectral domain feature extraction capability of GCN.
3) We design the neighborhood convolution to fuse local spectral information to prevent the features extracted by the Transformer from being too scattered and to fully exploit the local features in the spectral domain.

Complete experiments are conducted on three hyperspectral datasets to demonstrate the effectiveness and robustness of GTFN. Comparing 11 different HSI classification algorithms, GTFN achieves the highest classification accuracy.

II. Review of GCN and Transformer

This section briefly introduces the two HSI classification models by reviewing the definitions and basic representations of GCN and Transformer.

A. Graph Convolutional Network

The GCN can denote HSI images as undirected graphs, thus explicitly establishing relationships between different pixel points and modeling spatial pixel points. This provides spatial dependencies for classifying pixels, further improving the model classification performance. For a graph $G = (V, E, A)$, V denotes the set of vertices, $|V| = N$. E is the set of edges

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of the graph, $A \in R^{N\times N}$ is the adjacency matrix of the graph, and given the adjacency matrix A and the degree matrix D, the graph Laplacian operator is

$$L = D - A. \tag{1}$$

Normalizing the Laplace matrix above yields

$$L_n = I - D^{\frac{1}{2}} A D^{-\frac{1}{2}} \tag{2}$$

where I is the unit matrix and L_n is the corresponding symmetrically normalized Laplacian matrix.

To embed the denotation of the nodes, a convolution kernel $g_\theta = \mathrm{diag}(\theta)$ is introduced, where $\theta \in R^N$; the formula for the convolution of the graph can be expressed as

$$x * g_\theta = U g_\theta(\Lambda) U^T x \tag{3}$$

where x is the input signal, U is the eigenvector matrix of L_n, Λ is the eigenvalue of L_n, and g_θ can be considered as a function of the eigenvalue Λ. To simplify the calculation, a Chebyshev polynomial is used to fit $g_\theta(\Lambda)$

$$g_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta_K T_K(L_n) \tag{4}$$

where T_K is the Chebyshev polynomial.

Taking the first-order Chebyshev polynomial for further simplification gives

$$x * g_\theta = \theta\left(\tilde{D}^{\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}\right) x \tag{5}$$

where $\tilde{A} = A + I$ and $\tilde{D} = \sum_j \tilde{A}_{ij}$.

Based on (5), the propagation rule of GCN is given as follows:

$$H^{(l+1)} = \sigma\left(\tilde{D}^{\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)}\right) \tag{6}$$

where $H^{(l+1)}$ and $H^{(l)}$ represent the values of layer $l+1$ and layer l, respectively, σ represents the activation function, and W represents the weight matrix.

B. Vision Transformer

The spectral sequence in HSI can be seen as a sequence-based data structure, and Transformer is excellent at extracting sequence features and building a global attention mechanism. Transformer mainly consists of a multilayer perceptron (MLP) and an MHSA mechanism, as shown in Fig. 3.

Suppose that the input sequence is x and the length is m. First, the input vector p is obtained by embedding the position vector and x, and the sequence data are mapped into three vectors, query (Q), key (K), and value (V), by sharing the weights Wq, Wk, and Wv, respectively. Then, the SA layer can be expressed as

$$Z = \mathrm{Attention}(Q, K, V) = \mathrm{soft\,max}\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{7}$$

where d denotes the dimension of Q and K. Z is the output of the SA layer, and if there are N SA layers, Z can be denoted as Zi, $i = 1, 2, \ldots, N$. The output of the MHSA can be denoted as

$$Z_{\mathrm{MHSA}} = \mathrm{MHSA}(Z_i) = \mathrm{Concat}(Z_1, Z_2, \ldots, Z_N) W' \tag{8}$$

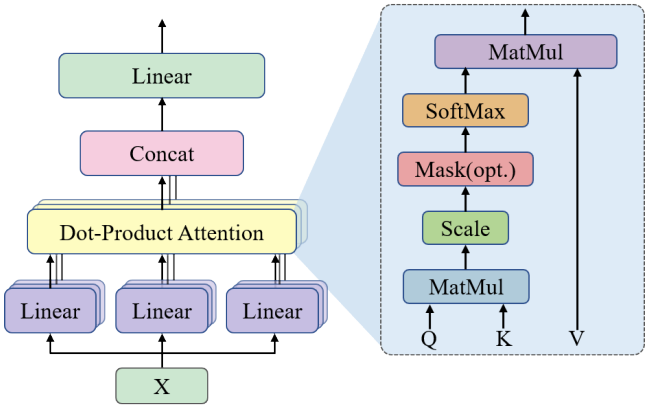


Fig. 3. Attention mechanism in Transformer. (Left) MHSA. (Right) SA.

where Z_{MHSA} is the output of the MHSA and W' is the projection matrix.

Transformer uses LayerNorm and residual connectivity in MHSA and MLP to enhance the model performance, so the encoding module in Transformer can be summarized as

$$Z' = \mathrm{MHSA}(\mathrm{LN}(p)) + p \tag{9}$$

$$Z = \mathrm{MLP}(\mathrm{LN}(Z')) + Z' \tag{10}$$

where LN stands for layer normalization and Z' and Z denote the output of the MHSA and MLP, respectively.

III. Proposed Methodology

In this section, the workflow and overall structure of GTFN are first described. Next, the Follow Patch for joint spatial context information is described in detail, followed by a detailed description of the spectral attention GCN. Finally, the workflow and specific structure of the local spectral feature fusion Transformer are elaborated.

A. Overview of GTFN

To deeply mine the information in the spatial-spectral domain of HSI, this article fully combines the spatial modeling capability of GCN for pixels and the feature extraction capability of Transformer for spectral sequences and proposes a new end-to-end classification framework GTFN for more fine-grained recognition and characterization of HSI images. GTFN mainly consists of two components, SAGCN and local spectral feature fusion Transformer (LST), and two modules are integrated into the network in a cascade manner. First, HSI is divided into multiple patches of the same size. To fully use the spatial neighborhood information of the classified pixels, the Follow Patch is designed to be different from the simple patch in ViT. At the same time, the input to SAGCN in the form of Follow Patches alleviates the problem of excessive consumption of computational resources due to the large adjacency matrix. Next, each Follow Patch adjacency matrix is built and input to the SAGCN. To fully incorporate the spatial-spectral domain information, a spectral attention module is introduced in GCN, making the GCN pay more attention to the essential spectral information. Then, SAGCN

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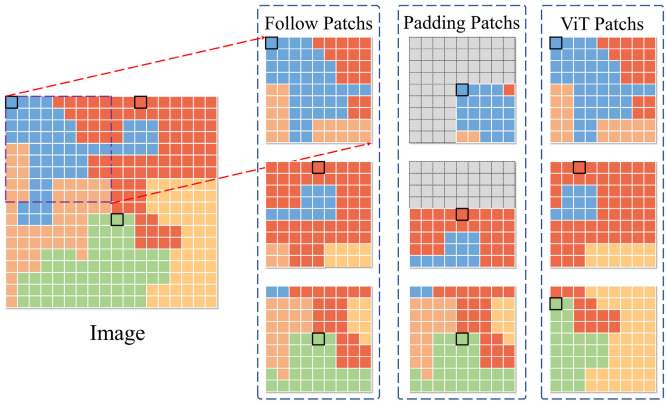


Fig. 4. Illustration of the Follow Patch. The input image is on the left, the black-bordered pixels are the target pixels, and the three blue dashed boxes on the right represent the Patchs of the target pixels obtained by the three methods, which show that the same patch size allows this method to make better use of the surrounding pixel information.

establishes the spatial relationship of pixels, outputs the feature map, selects the classification features in the feature map through pixels selection block, and uses them as the input of LST. Finally, this article designs a neighborhood convolution to fuse the local spectral domain information of the LST input features, and the LST extracts the spectral domain information and obtains the final classification accuracy by a linear classifier.

B. Follow Patch

Due to the large number of pixels in HSI, it would be computationally intensive to consider each pixel as a node for input into the GCN network. Therefore, Wan et al. [13] performed superpixel segmentation of the whole image and treated each superpixel as a node to reduce the computational complexity. However, inputting all the samples into the network at once would cause a large amount of memory usage, and the gradient decreases more slowly. The miniGCN proposed in [11] uses a small sample sampler to divide the graph into multiple subgraphs and compute the node features for each sample separately and then sums up each subgraph node to obtain the global node feature information. Both methods have certain drawbacks: the method based on superpixel segmentation is not flexible enough and has serious memory consumption; the miniGCN random sampling approach is less efficient and the process is more cumbersome. To address the above problems, this article designs a Follow Patch as the input to the GCN network, as shown in Fig. 4.

The intention of Follow Patch is to reduce the computational complexity and training complexity while exploiting the local spatial information of the classification nodes. Assuming that the classification pixel is $x(x_{\text{center}}, y_{\text{center}})$ (marked by the black box in Fig. 4), a patch of size $m \times m$ around x is delineated and fed into the SAGCN network as a classification unit of one pixel.

1) Since the patch is a square region, the position of the patch in the image can be determined by calibrating the coordinates of the top-left C_l and the bottom-right position C_r of the patch.

2) The coordinates of C_l of Follow Patch are (x_p, y_p), here with m as an odd number

$$x_p = \begin{cases} x_{\text{center}} - \dfrac{m-1}{2}, & x_{\text{center}} > \dfrac{m-1}{2} \text{ and} \\ & x_{\text{center}} < W - \dfrac{m-1}{2} \\ 0, & x_{\text{center}} \le \dfrac{m-1}{2} \\ W - m - 1, & x_{\text{center}} \ge W - \dfrac{m-1}{2} \end{cases} \tag{11}$$

$$y_p = \begin{cases} y_{\text{center}} - \dfrac{m-1}{2}, & y_{\text{center}} > \dfrac{m-1}{2} \text{ and} \\ & y_{\text{center}} < H - \dfrac{m-1}{2} \\ 0, & y_{\text{center}} \le \dfrac{m-1}{2} \\ H - m - 1, & y_{\text{center}} \ge H - \dfrac{m-1}{2}. \end{cases} \tag{12}$$

The C_r coordinates (x_q, y_q) are calculated as follows:

$$x_q = \begin{cases} x_{\text{center}} + \dfrac{m-1}{2}, & x_{\text{center}} > \dfrac{m-1}{2} \text{ and} \\ & x_{\text{center}} < W - \dfrac{m-1}{2} \\ m - 1, & x_{\text{center}} \le \dfrac{m-1}{2} \\ W - 1, & x_{\text{center}} \ge W - \dfrac{m-1}{2} \end{cases} \tag{13}$$

$$y_q = \begin{cases} y_{\text{center}} + \dfrac{m-1}{2}, & y_{\text{center}} > \dfrac{m-1}{2} \text{ and} \\ & y_{\text{center}} < H - \dfrac{m-1}{2} \\ m - 1, & y_{\text{center}} \le \dfrac{m-1}{2} \\ H - 1, & y_{\text{center}} \ge H - \dfrac{m-1}{2} \end{cases} \tag{14}$$

where W and H are the width and height of the image, respectively.

Unlike CNN, where the convolution region is selected in the form of padding, the Follow Patch extends the spatial information of the image edge points, enabling the acquisition of richer spatial features. In addition, the Follow Patch avoids the problem of classification pixel marginalization compared with the patch division in ViT.

The spatial information of classified pixel points in HSIs is often only closely related to the surrounding features. Compared with building global node information, the Follow Patch removes redundant spatial information in the form of aggregated local spatial information, making classification and computation more efficient.

C. Spectral Attention GCN

The GCN model establishes spatial dependencies between different nodes. However, the correlation between different

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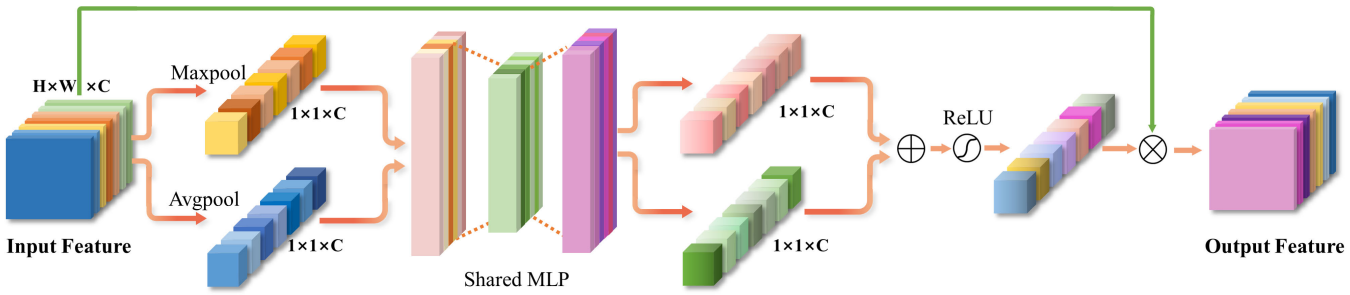


Fig. 5. Illustration of the spectral attention mechanism module.

spectral segments is neglected. In this article, the SAGCN module is designed to improve the classification accuracy by adding a spectral attention module to highlight the spectral information that is more valuable for classification, suppresses redundant information, and achieves a fusion of information in the spatial-spectral domain.

The spectral attention mechanism is shown in Fig. 5, where feature maps' spatial information is aggregated through averaging pooling and max pooling to generate two disparate feature maps with spatial contextual information descriptions. These two feature maps are then passed through a shared fully connected layer to generate a frequency domain attention feature map. The attention feature maps are summed and then passed through an activation function to aggregate the attention information of the two feature maps to obtain spectral attention weights.

Assuming that the size of the GCN output feature map F is $H \times W \times C$, the spectral attention weight S of size $1 \times 1 \times C$ is formed through the attention mechanism, and the feature map F is multiplied by the weight S to obtain the spectral attention machine features.

The calculation of the weight coefficients can be defined as

$$\begin{aligned} S &= M_c(F) \\ &= f(\mathrm{MLP}(\mathrm{AvgPool}(F)) + \mathrm{MLP}(\mathrm{MaxPool}(F))) \\ &= \sigma\left(W_1\left(W_0\left(F^c_{\mathrm{avg}}\right)\right) + W_1\left(W_0\left(F^c_{\mathrm{max}}\right)\right)\right) \end{aligned} \tag{15}$$

$$F^c_{\mathrm{avg}} = \frac{1}{H \times W}\sum_{m=1}^{H}\sum_{n=1}^{W} F(m,n), \quad F^c_{\mathrm{max}} = \max(F) \tag{16}$$

where F^c_{avg} is the feature map of F after average pooling, F^c_{max} is the feature map of F after maximum pooling, $F(m,n)$ represents the value at (m,n) on the feature map, MaxPool denotes the maximum pooling operation, AvgPool denotes the average pooling operation, and MLP denotes the shared multilayer neural network. f represents the activation function, σ is the coefficient, and W_1 and W_0 are the parameters of the MLP.

S is multiplied one by one with the spectral band features to obtain the spectral domain attention output F'

$$F' = S \otimes F \tag{17}$$

where $\otimes$ represents the weights, and each spectral domain feature is multiplied one by one.

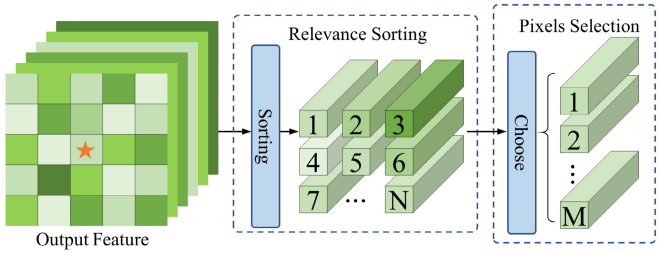


Fig. 6. Illustration of the pixel selection module. On the left side is the output feature map, the marked red point is the classified pixel, the correlation is sorted according to the weights of classified pixels and other nodes in the adjacency matrix, and the top M points after sorting are selected as the chosen pixel.

D. Local Spectral Feature Fusion Transformer

When using ViT for classification, the image is usually segmented into nonoverlapping patches using the patch embedding module, whereas in the HSI classification, the classification is done for each pixel. Therefore, the Transformer input is usually a single pixel, and in order to exploit the full spatial information of the HSI, a patch-wise approach to input is used in [8], which takes the classification pixel and several surrounding pixels as a patch input. This approach improves the classification capability compared with individual pixel inputs, but the indiscriminate input of surrounding pixels is highly susceptible to misclassification for feature edge pixels.

In this article, we take advantage of the GCN to select the pixel with the strongest correlation with the current classification pixel to form the input of Transformer, as shown in Fig. 6. The spatial information can be used to establish spatial remote dependencies, avoiding the problem of misclassification of edge pixels and further reducing the computational complexity.

Assuming that the classification nodes are V_i, the nodes are first sorted according to the weights of the nodes of the adjacency matrix obtained in the GCN

$$S^i_j = \mathrm{Sort}(\mathrm{Corr}(V_i, V_j)), \quad i, j = 1, 2, \ldots, N \tag{18}$$

$$V_{ij} = \mathrm{Corr}(V_i, V_j) \tag{19}$$

where V_{ij} denotes the weight between nodes V_i and V_j, the Sort function is used to sort all nodes V_j by V_{ij} from largest to smallest, and N denotes the number of all nodes.

The first M points of S^i_j are taken to form the input of the Transformer

$$F^i_{\mathrm{in}} = \mathrm{Concat}(S_1, S_2, \ldots, S_j, \ldots, S_M), \quad j = 1, 2, \ldots, M \quad (M \le N) \tag{20}$$

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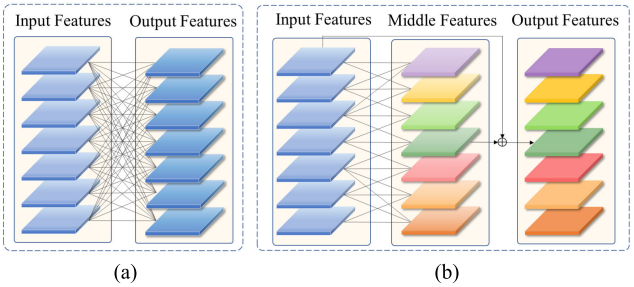


Fig. 7. Illustration of neighborhood convolution. The left side is normal 2-D convolution. The right side is neighborhood convolution. (a) Normal 2-D convolution. (b) Neighborhood convolution.

where Concat represents the splicing operation and the top M selected nodes that are most relevant to the nodes in V_i are formed into a new input F_{in}^i.

Compared with directly using the whole patch as input, the relevance ranking method is more advantageous for GCN to model spatial nodes and extract the classification nodes that are more relevant to the current node, thus removing distracting information to avoid misclassification while reducing the computational complexity and saving computational resources. In addition, the features obtained through the pixel selection module as input to Transformer are more suitable for Transformer processing, and the problem of identical patches for different classification nodes in the Follow Patch is well solved.

Transformer can establish remote dependencies of spectral sequences but often needs to focus better on local spectral information. Each pixel of HSI can be seen as approximately continuous spectral information, and the subtle differences of adjacent spectral bands can distinguish feature information well. While using Transformer to establish global attention and to better fuse local information, this article creatively proposed neighborhood convolution, as shown in Fig. 7, first fusing the features of the current and neighboring spectral bands and then combining the original spectral information using the form of residual concatenation as the final output.

First, the neighboring bands of the ith spectrum X_i were extracted

$$X_{\text{group}}^i = \text{Concat}(X_m, X_{m+1}, \ldots, X_i, \ldots, X_{m+k-1}) \tag{21}$$

$$\begin{cases} m = i - \dfrac{k-1}{2}, & i > \dfrac{k-1}{2} \text{ and } i < N - \dfrac{k-1}{2} \\ m = 0, & i \le \dfrac{k-1}{2} \\ m = N - k - 1, & i \ge N - \dfrac{k-1}{2} \end{cases} \tag{22}$$

where X_{group}^i is the neighboring spectrum of the ith spectrum, k is the number of neighboring spectrum, and N is the total number of spectrums.

Convolution of the neighborhood of X_{group}^i and joining the residuals to the original spectrum

$$X_i' = W_i X_{\text{group}}^i + b_i, i = 1, 2, \ldots, N \tag{23}$$

$$X_i'' = X_i' + X_i \tag{24}$$

$$F = \text{Concat}\left(X_1'', X_2'', \ldots, X_N''\right) \tag{25}$$

where W_i is the weight matrix of the ith neighborhood convolution and b_i is the bias of the ith neighborhood convolution to obtain the neighbor channel feature X_i'; then, the output feature X'' is obtained after residual concatenation, and finally, X'' is cascaded to obtain the output feature map F.

Neighborhood convolution allows better aggregation of local information than fully connected layers while significantly reducing the computational complexity. To prevent overfitting due to fewer parameters, the BN layer and ReLU function are used to regularize the neighborhood convolution to improve its classification performance

$$\begin{aligned} F' &= \text{ReLU}(\text{BN}(F)) \\ &= \text{ReLU}\left(\gamma \frac{F-\mu}{\sigma} + \beta\right) \\ &= \max\left(0, \gamma \frac{F-\mu}{\sigma} + \beta\right) \end{aligned} \tag{26}$$

where μ and σ represent the expectation and variance of a batch, respectively, γ and β are parameters that can be learned during training, and F' is the output after the regularization of the feature layer F.

IV. Experiment

In this section, three well-known hyperspectral datasets and experimental setups are first described. The effectiveness of this algorithm is then demonstrated by comparing it with one classical algorithm and ten DL models. Finally, the network structure's effectiveness and the hyperparameter settings' effect on the experimental results are analyzed through ablation experiments.

A. Data Description

Indian Pines (IP) dataset is acquired by the AVIRIS sensor over Northwestern Indiana and contains 224 bands and 145 × 145 pixels. After removing the interference bands, the remaining 200 bands are selected for 16 features and 10 249 marked feature pixels. In the experiments, 435 samples from the labeled features are selected as the training set and 9814 samples as the test set. The selection of samples for each category is specified in Table I.

Salina (SA) dataset is captured by the AVIRIS sensor over the Salinas Valley, California. The coverage area consists of 512 × 217 samples and 224 bands, leaving 204 bands after removing the interfering bands. It includes 16 feature types with a total of 111 104 pixels, of which 54 129 are classified pixels and 56 975 are background pixels. In the experiments, 480 samples from the labeled features are selected as the training set, and 53 649 samples are selected as the test set; the selection of samples for each category is specified in Table II.

Pavia University (UP) dataset is captured over Pavia University and contains 610 × 340 pixels and 115 bands. After removing the interfering bands (low signal-to-noise ratio bands and so on), the remaining 103 bands are selected, with a total of nine types of features and 42 776 marked feature pixels. In the experiments, 270 samples from the labeled features are

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TABLE I
CATEGORY INFORMATION FOR IP DATASET

ClassID	Class name	Training	Testing
1	Alfalfa	15	31
2	Corn-notill	30	1398
3	Corn-mintill	30	800
4	Corn	30	207
5	Grass-pasture	30	453
6	Grass-trees	30	700
7	Grass-pasture-mowed	15	13
8	Hay-windrowed	30	448
9	Oats	15	5
10	Soybeans-notill	30	942
11	Soybean-mintill	30	2425
12	Soybean-clean	30	563
13	Wheat	30	175
14	Woods	30	1235
15	Buildings-grass-trees-drivers	30	356
16	Stone-steel-towers	30	63
	Total	435	9814

TABLE II
CATEGORY INFORMATION FOR SA DATASET

ClassID	Class name	Training	Testing
1	Brocoli_green_weeds_1	30	1979
2	Brocoli_green_weeds_2	30	3696
3	Fallow	30	1946
4	Fallow_rough_plow	30	1364
5	Fallow_smooth	30	2648
6	Stubble	30	3929
7	Celery	30	3549
8	Grapes_untrained	30	11241
9	Soil_vinyard_develop	30	6173
10	Corn_senesced_green_weeds	30	3248
11	Lettuce_romaine_4wk	30	1038
12	Lettuce_romaine_5wk	30	1897
13	Lettuce_romaine_6wk	30	886
14	Lettuce_romaine_7wk	30	1040
15	Vinyard_untrained	30	7238
16	Vinyard_vertical_trellis	30	1777
	Total	480	53649

selected as the training set and 42 506 samples as the test set, and the selection of samples for each category is specified in Table III.

B. Experiment Settings

1) Evaluation Metrics: Overall classification accuracy (OA), average classification accuracy (AA), and Kappa coefficient, which is a metric used to evaluate the degree of consistency of judgments, are used as quantitative evaluation metrics throughout the overall. The OA represents the overall classification accuracy. In addition, qualitative analysis is carried out by visualizing the classification results.

2) Comparison With the State-of-the-Art Backbone: In this article, GTFN is compared with one selected classical method and ten DL methods with GTFN to analyze

TABLE III
CATEGORY INFORMATION FOR UP DATASET

ClassID	Class name	Training	Testing
1	Asphalt	30	6601
2	Meadows	30	18619
3	Gravel	30	2069
4	Trees	30	3034
5	Metal Sheets	30	1315
6	Bare-soil	30	4999
7	Bitumen	30	1300
8	Bricks	30	3652
9	Shadows	30	917
	Total	270	42506

algorithm effectiveness comprehensively. The methods are the SVM-RBF, the 3-D CNN, the DFFN [19], the RSSAN, the DBDA, the MDGCN, the WFCG, the AMGCFN, the Spectralformer (SF) [8], the SSFTT [9], and the GAHT.

3) Parameter Settings and Implementation Details: All experiments were conducted on NVIDIA Titan RTX using the Pytorch framework. For a fair comparison, in both the comparison experiments and GTFN, the patch size is set to 9×9, the epoch is 200, the batch size is 64, the training optimizer is the Adam optimizer, and the learning rate is 5×10^{-4}. The specific network structures have been adopted from the proposals in the original paper.

For GTFN, the Follow Patches are fed into a four-stage cascaded GCN network, and the last layer of the GCN uses the spectral attention mechanism. The number of pixels selected by the pixel selection module is 19. Five cascaded coding modules are used for the Transformer part, each consisting of four SA heads, eight MLPs, and the GELU nonlinear activation layer.

C. Contrast Experiment Results and Analysis

This section compares GTFN with traditional algorithms and the most widely used and advanced DL algorithms.

1) Comparative Experimental Analysis: Tables IV–VI show the quantitative classification results and each category's classification accuracy for the above 12 classification algorithms on the IP, UP, and SA datasets. Overall, the classification accuracy of the traditional algorithm SVM-RBF is lower than the other DL methods in terms of OA, AA, and Kappa. The CNN-based classification methods, such as 3-D CNN, DFFN, RSSAN, and DBDA, generally show good classification performance. There is a certain gap in classification accuracy compared with the Transformer framework-based classification networks SF, SSFTT, GAHT, and GTFN. The GCN-based classification method MDGCN and CNN and GCN fusion-based methods WFCG and AMGCFN also have remarkable classification performance. The GCN and CNN fusion-based methods achieve excellent classification performance by fully utilizing the advantages of CNN capturing pixel-level information and GCN aggregating contextual information. Since the Transformer framework has a strong ability to handle sequence features, which can well mine the spectral

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TABLE IV
QUANTITATIVE COMPARISON OF IP DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	SVM-RBF	3D CNN	DFFN	RSSAN	DBDA	MDGCN	WFCG	AMGCFN	SF	SSFTT	GAHT	GTFN
1	31.36±3.65	87.01±0.00	96.77±2.63	31.18±25.85	100.00±0.00	97.91±2.94	82.33±3.50	100.00±0.00	47.31±13.52	100.00±0.00	98.92±1.52	**100.00±0.00**
2	58.16±4.99	41.18±6.81	72.22±2.52	60.94±13.57	83.69±0.44	74.61±9.02	85.99±3.11	83.08±10.17	50.74±5.01	77.23±2.92	80.31±2.26	**90.30±0.58**
3	51.12±5.52	46.92±8.72	91.46±3.72	64.75±9.29	89.87±0.00	84.39±3.21	79.80±2.77	88.87±7.21	57.04±6.71	77.46±10.09	84.83±7.30	**91.79±0.46**
4	30.90±1.75	88.89±2.40	97.58±1.18	90.02±4.96	90.98±0.99	97.42±1.94	75.29±2.59	**100.00±0.00**	84.06±6.35	98.39±1.39	99.03±1.04	98.71±1.21
5	83.18±0.69	73.36±6.93	94.11±1.38	83.81±5.89	93.23±0.99	95.05±0.87	**99.34±0.76**	95.12±5.01	72.33±6.60	93.75±4.06	91.69±3.97	94.19±0.38
6	91.13±1.14	88.29±4.15	96.14±2.02	90.95±2.90	97.61±0.37	**99.42±0.11**	98.68±0.73	97.37±1.33	82.29±0.91	96.67±1.36	96.62±1.35	96.71±1.05
7	44.18±10.37	100.00±0.00	100.00±0.00	100.00±0.00	94.87±3.62	100.00±0.00	72.02±17.02	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**	94.87±3.63
8	95.79±1.35	97.02±1.00	97.69±0.69	99.26±0.59	**100.00±0.00**	100.00±0.00	99.48±0.27	99.77±0.18	96.35±1.24	99.78±0.32	99.85±0.21	99.85±0.11
9	18.17±1.37	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	66.13±7.48	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
10	56.76±2.69	57.93±3.73	88.46±0.30	78.91±8.61	88.39±1.13	91.63±2.59	69.28±4.05	83.24±6.53	71.09±4.68	87.30±2.73	86.13±3.30	**96.04±1.21**
11	73.59±3.41	59.64±3.73	68.14±1.90	56.78±8.61	75.79±0.76	75.21±4.71	**93.44±2.42**	90.03±6.39	53.46±9.35	67.68±7.54	75.77±4.82	90.38±0.72
12	46.22±5.87	62.94±5.66	87.63±1.48	65.78±4.34	89.16±0.88	**93.30±2.19**	82.08±8.35	90.98±5.12	43.64±4.48	83.36±2.76	88.93±2.04	90.70±1.91
13	86.97±0.51	99.05±0.27	99.62±0.54	98.48±0.97	**100.00±0.00**	100.00±0.00	98.89±1.56	99.61±0.53	97.71±1.62	99.62±0.54	99.43±0.81	99.62±0.54
14	94.43±0.82	83.13±0.60	92.52±0.28	96.06±0.42	89.58±0.03	94.67±1.64	97.64±0.65	97.18±0.47	76.60±5.15	91.77±1.61	90.31±0.83	**99.43±0.30**
15	51.70±2.47	80.43±6.91	89.51±3.78	79.78±7.24	98.59±0.39	**99.90±0.13**	82.96±5.19	98.77±0.81	85.49±3.32	95.51±3.58	95.51±3.39	99.16±0.00
16	83.17±8.52	100.00±0.00	99.47±0.75	98.41±1.30	98.41±0.00	100.00±0.00	94.53±2.37	99.47±0.75	98.94±1.50	96.30±5.24	**100.00±0.00**	99.47±0.75
OA (%)	66.96±2.25	65.83±1.78	83.61±1.01	73.78±1.04	87.06±0.33	86.77±2.46	87.68±0.24	91.25±4.84	65.24±2.09	82.96±1.21	85.96±1.69	**94.00±0.11**
AA (%)	62.30±1.66	79.12±1.21	91.96±0.70	80.94±1.22	93.13±0.08	93.97±0.78	86.12±1.40	95.21±5.47	76.07±0.95	91.55±0.97	92.96±0.79	**96.33±0.26**
Kappa	62.81±2.42	61.38±1.99	81.47±1.15	70.48±0.99	85.32±0.38	85.03±2.73	86.03±0.26	90.03±5.47	61.06±2.1	80.71±1.29	84.05±1.88	**93.15±0.13**

TABLE V
QUANTITATIVE COMPARISON OF SA DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	SVM-RBF	3D CNN	DFFN	RSSAN	DBDA	MDGCN	WFCG	AMGCFN	SF	SSFTT	GAHT	GTFN
1	98.70±1.05	98.77±0.80	94.81±4.37	98.84±0.25	99.86±0.10	97.82±3.07	99.74±0.35	100.00±0.00	94.74±0.93	100.00±0.00	100.00±0.00	**100.00±0.00**
2	99.20±0.55	98.70±0.71	99.25±0.12	99.50±0.06	99.91±0.11	100.00±0.00	**100.00±0.00**	99.72±0.48	98.82±0.49	99.49±0.46	99.98±0.03	99.97±0.04
3	92.85±0.61	90.00±1.19	95.89±2.40	97.14±1.82	98.33±0.63	100.00±0.00	**100.00±0.00**	99.95±0.09	92.17±1.99	90.10±7.84	98.49±0.96	99.97±0.04
4	97.54±0.17	98.75±0.36	96.51±1.00	99.49±0.41	97.80±0.73	**99.85±0.10**	98.99±0.38	98.53±2.23	96.26±0.94	99.76±0.30	99.12±0.75	99.59±0.19
5	97.65±0.65	95.20±1.07	97.52±0.69	96.99±1.21	98.74±0.71	97.09±0.00	97.87±0.71	97.25±0.73	89.41±1.83	97.13±2.30	**99.13±0.44**	95.89±1.36
6	99.89±0.02	99.40±0.39	99.47±0.73	99.02±0.66	**99.99±0.01**	99.92±0.00	99.96±0.04	99.59±0.63	99.20±0.72	99.71±0.31	99.97±0.04	99.84±0.09
7	98.76±0.43	99.38±0.37	99.49±0.30	98.07±1.66	99.84±0.14	92.12±3.16	99.80±0.27	**99.99±0.02**	97.91±1.99	99.03±0.50	99.80±0.16	99.72±0.26
8	77.68±1.55	72.55±6.82	79.11±2.26	71.60±1.74	85.29±0.44	91.97±3.06	77.52±7.72	83.51±9.23	73.07±1.12	73.95±3.67	83.11±3.18	**92.18±0.80**
9	99.18±0.24	93.30±0.93	94.76±0.58	98.15±0.96	99.54±0.27	**100.00±0.00**	99.90±0.11	99.96±0.07	93.77±1.42	98.59±1.35	99.47±0.11	99.80±0.20
10	83.99±1.33	88.24±0.95	91.81±0.85	92.03±2.89	94.42±2.87	83.65±1.69	94.71±2.08	**99.61±0.34**	91.39±1.83	94.28±1.14	97.36±1.81	95.42±0.53
11	89.42±0.31	93.96±2.26	98.75±0.75	96.76±1.87	97.54±2.98	96.91±0.00	98.74±0.56	99.33±0.59	92.36±3.31	98.65±0.27	98.62±1.14	**99.77±0.22**
12	95.31±0.35	99.33±0.22	99.95±0.07	99.37±0.75	98.47±0.06	89.10±4.05	100.00±0.00	99.91±0.15	99.05±0.58	99.95±0.04	**100.00±0.00**	99.66±0.21
13	94.17±1.21	99.06±0.93	**100.00±0.00**	99.44±0.65	99.06±0.73	96.95±0.00	99.96±0.05	99.85±0.13	99.40±0.43	97.86±1.28	99.96±0.05	99.50±0.43
14	90.11±4.68	96.73±0.95	**99.26±0.32**	95.54±2.70	98.59±0.55	96.98±0.18	98.04±0.76	99.23±0.39	96.60±0.95	98.91±0.16	98.94±0.21	99.17±0.87
15	60.59±1.96	71.35±3.32	83.37±0.37	79.07±3.13	71.66±2.15	71.74±2.79	84.93±7.15	89.77±10.32	81.58±2.20	81.10±3.63	87.10±2.83	**93.06±0.65**
16	97.04±1.20	92.61±1.97	93.60±1.91	95.72±2.55	99.19±0.51	**100.00±0.00**	99.15±0.45	99.90±0.17	93.72±2.04	96.96±2.05	96.83±2.66	99.18±0.31
OA (%)	87.31±0.53	87.59±1.49	91.35±0.41	89.66±0.76	92.18±0.25	92.20±0.77	92.68±1.07	94.87±1.07	88.80±0.21	90.67±1.52	94.21±0.39	**96.81±0.23**
AA (%)	92.01±0.15	92.96±0.56	95.22±0.13	94.79±0.71	96.14±0.12	94.63±0.42	96.83±0.40	97.88±0.50	93.09±0.19	95.34±1.11	97.37±0.07	**98.30±0.09**
Kappa	85.90±0.58	86.21±1.64	90.39±0.45`	88.53±0.83	91.30±0.27	91.30±0.86	91.86±1.18	94.30±1.19	87.57±0.24	89.64±1.68	93.56±0.43	**96.45±0.25**

TABLE VI
QUANTITATIVE COMPARISON OF UP DATASET, WITH BEST PERFORMANCE IN BOLD

Class No.	SVM-RBF	3D CNN	DFFN	RSSAN	DBDA	MDGCN	WFCG	AMGCFN	SF	SSFTT	GAHT	GTFN
1	92.98±2.74	80.72±4.59	87.55±8.07	77.98±10.08	92.81±0.94	88.96±2.94	**98.79±0.62**	88.21±3.49	72.19±6.62	87.56±3.49	92.12±3.55	89.62±0.72
2	92.11±1.14	73.15±6.14	71.22±4.63	87.64±7.48	92.82±0.37	90.73±2.10	**98.31±0.48**	93.98±1.10	75.51±3.03	92.35±6.27	91.19±5.60	97.24±0.47
3	60.58±5.69	71.61±2.95	77.57±13.44	59.55±6.98	83.72±0.83	84.98±6.85	80.41±8.28	**94.55±2.09**	75.17±8.81	86.37±2.63	87.26±4.99	84.52±3.29
4	78.11±7.24	97.25±0.82	92.17±1.65	96.84±0.49	92.66±0.30	97.40±0.51	94.74±3.44	89.85±4.29	89.53±2.28	**97.90±0.43**	97.73±0.13	94.91±0.46
5	91.29±5.14	99.92±0.11	99.97±0.04	99.34±0.48	98.88±0.14	99.39±0.93	99.89±0.09	100.00±0.00	100.00±0.00	99.97±0.04	**100.00±0.00**	98.71±0.35
6	55.21±2.88	74.67±6.94	71.07±8.66	68.59±7.98	92.44±0.20	93.88±0.92	80.61±11.35	97.93±1.70	76.65±0.60	87.62±2.06	96.03±3.41	**99.59±0.17**
7	52.08±7.58	86.15±4.74	81.95±3.62	86.33±7.68	87.79±0.50	**99.59±0.65**	85.77±5.24	96.22±4.12	70.21±6.54	92.64±3.16	95.36±4.66	98.97±0.42
8	82.62±9.97	72.32±1.69	78.55±5.40	74.10±11.02	90.53±0.39	**94.68±7.64**	88.81±0.24	89.65±2.31	62.05±7.83	86.73±3.03	91.65±0.18	91.59±2.65
9	99.89±0.08	99.27±0.66	99.64±0.22	99.64±0.37	96.18±0.31	99.85±0.17	98.61±0.98	91.53±4.70	97.89±1.22	**99.93±0.10**	99.82±0.10	96.07±0.70
OA (%)	79.74±0.94	77.87±1.37	78.00±0.86	82.61±2.28	92.23±0.25	92.10±0.78	92.98±2.19	93.12±0.61	76.03±0.79	91.08±2.61	92.81±2.32	**95.14±0.18**
AA (%)	78.32±0.30	83.90±0.62	84.41±2.51	83.33±2.41	91.98±0.10	94.38±0.28	91.77±1.27	93.55±0.91	79.91±0.83	92.34±0.92	94.57±0.40	**94.58±0.50**
Kappa	73.97±1.08	71.82±1.38	71.92±1.29	77.19±2.92	89.80±0.32	89.68±1.00	90.83±2.76	90.96±0.78	69.35±0.89	88.32±3.26	90.64±2.89`	**93.58±0.24**

features of HSI and form a global attention mechanism, classifiers based on the Transformer framework can rival or surpass CNN-based or GCN-based classification models.

Although the Transformer-based classifier can capture the global information of the spectral sequence, its limited feature extraction capability, such as spatial modeling capability and local spectral domain feature extraction, leads to its performance bottleneck. To solve the above problems, the GTFN proposed in this article uses GCN to extract spatial contextual information and designs neighborhood convolution to fully fuse the spectral information of adjacent spectral bands, which greatly improves the classification performance. Specifically, GTFN improves OA, AA, and Kappa by 2.75%, 1.12%, and 3.12%, respectively, over AMGCFN on the IP dataset. In addition, GTFN has a very small standard deviation of classification metrics, indicating the strong stability of the algorithm. On the SA dataset, GTFN improved by 1.94%, 0.42%, and 2.15% over AMGCFN's OA, AA, and Kappa, respectively. On the UP dataset, GTFN improved by 2.02%, 1.03%, and 2.62% over AMGCFN's OA, AA, and Kappa, respectively. It can be seen that GTFN is ahead of other models in all classification metrics, meanwhile, the GCN

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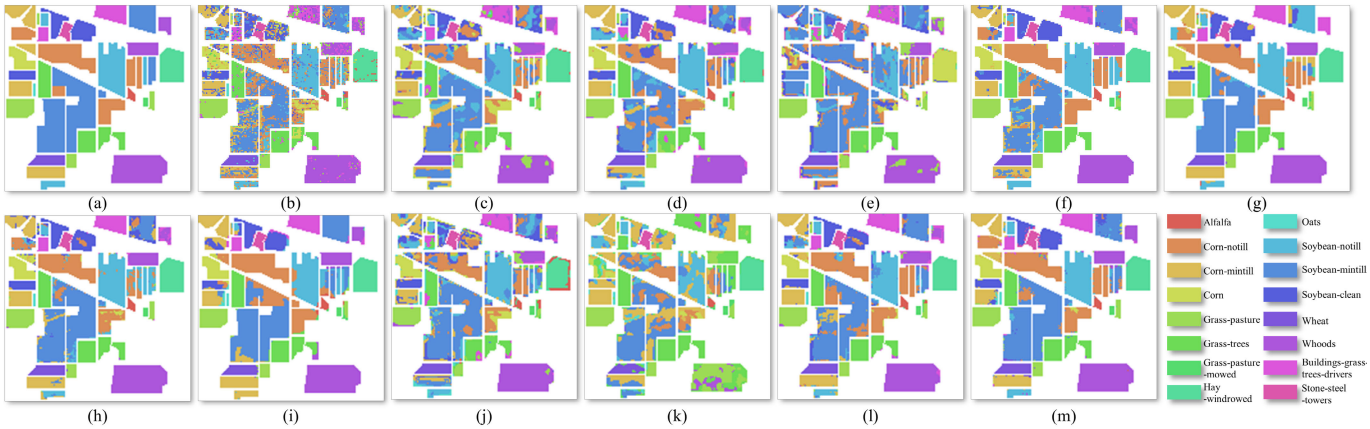


Fig. 8. Classification graph of IP dataset. (a) Ground Truth. (b) SVM-RBF. (c) 3-D CNN. (d) DFFN. (e) RSSAN. (f) DBDA. (g) MDGCN. (h) WFCG. (i) AMGCFN. (j) SF. (k) SSFTT. (l) GAHT. (m) GTFN.

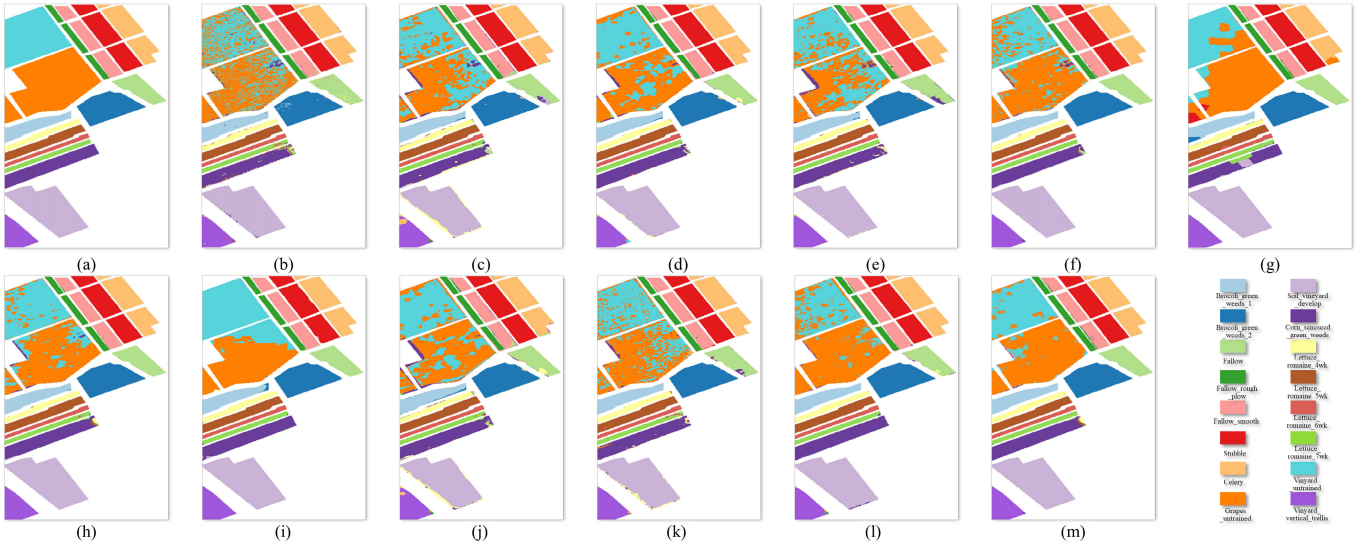


Fig. 9. Classification graph of SA dataset. (a) Ground Truth. (b) SVM-RBF. (c) 3-D CNN. (d) DFFN. (e) RSSAN. (f) DBDA. (g) MDGCN. (h) WFCG. (i) AMGCFN. (j) SF. (k) SSFTT. (l) GAHT. (m) GTFN.

and Transformer-based spatial-spectral domain feature mining classification.

2) Visual Analysis of Classification Results: In this article, the classification graphs of different classification methods are visualized to visualize the classification effect and to conduct qualitative analysis. Figs. 8–10 show the classification results of the IP dataset, UP dataset, and SA dataset, respectively. In the three datasets, the traditional classification method SVM-RBF does not work well, and pepper noise appears, which indicates that this method fails to identify the material of the feature well. In contrast, the DL-based method classifies better and obtains a smoother classification map, demonstrating CNNs' powerful nonlinear representation capability.

However, the local convolutional nature of CNN tends to cause inaccurate recognition of edge features, resulting in misclassification. Although the CNN and GCN fusion-based methods have better classification performance than the CNN-based methods, there are significant noise points on small surface areas. The Transformer-based classification framework is able to establish long-range dependencies in the spectral domain of an image. Therefore, the classification results are comparable to the abovementioned networks, alleviating the problem of oversmoothing of edge features. The GTFN combines the advantages of GCN and Transformer. On the one hand, GTFN can fully use the spatial contextual information of classification points and establish neighborhood relationships. On the other hand, it can model the spectral domain information and fuse the local spectral domain information to obtain a good classification mapping. It can be seen from the classification map that GTFN can show the texture and detail features of the features better than other methods, reduce the classification noise, and have stronger classification ability for edge pixels.

In addition, the network output features are transformed into a 2-D space by PCA to compare better the separability of the different features in this article, as shown in Fig. 11. The separability of features is better for GTFN in the three datasets. Specifically, the features are more dispersed for the SSFTT and GAHT, the feature boundaries are not clear, and the similar feature points are not compact. The GTFN output features overlap less, and the intraclass features are more compact and distributed more regularly. Therefore, based on the 2-D feature

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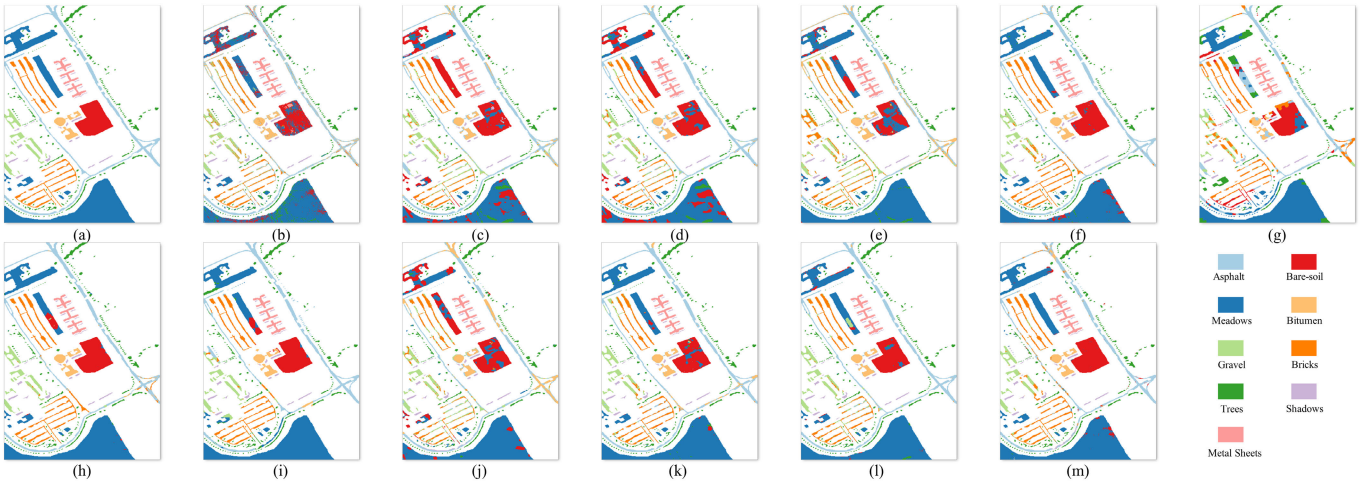


Fig. 10. Classification graph of UP dataset. (a) Ground Truth. (b) SVM-RBF. (c) 3-D CNN. (d) DFFN. (e) RSSAN. (f) DBDA. (g) MDGCN. (h) WFCG. (i) AMGCFN (j) SF. (k) SSFTT. (l) GAHT. (m) GTFN.

scatter plot, it can be seen that the GTFN output features are less distant within classes and more separable between classes and therefore have higher classification accuracy.

3) Training Samples Analysis: To further verify the effectiveness and robustness of GTFN, the OA of the 12 different algorithms was verified using different proportions of training samples, as shown in Fig. 12 (where the red dashes represent the OA values of GTFN). The training samples for the IP dataset were taken as 1%, 2%, 3%, 4%, and 5% of the dataset, and the training samples for the UP and SA datasets were taken as 0.25%, 0.5%, 1%, 2%, and 3%. From Fig. 12, it can be seen that as the training samples increase, the model is more adequately trained, and the OA gradually improves. Also, among all the algorithms, GTFN shows the best classification performance (the red fold is at the top of all the folds), with OA steadily improving as the proportion of samples increases. Compared with other algorithms, GTFN takes full account of spatial contextual information and establishes a global attention mechanism in the spectral domain at the same time and has the best classification accuracy for different datasets with different numbers of training samples. Therefore, GTFN has a high classification accuracy and strong generalization ability.

D. Ablation Experiment Setup and Model Analysis

1) Ablation Experiments: The spectral domain feature extraction capability of the GCN network is improved by introducing the spectral attention mechanism in GTFN, while the neighborhood convolution module is proposed to enhance the attention of Transformer's local spectral domain information. To further explore the gain of the two modules on the classification performance of GTFN, this article designs ablation experiments on the IP dataset to verify the effectiveness of the two modules for HSI classification. The results are shown in Table VII. Specifically, the GTFN classification framework can still achieve good classification results even without the neighborhood convolution and spectral attention mechanism, which demonstrates to a certain extent the potential of the proposed GCN and Transformer fusion network for HSI classification. In addition, adding either the spectral attention module or neighborhood convolution alone improves the classification results of the images, and the combination of the two yields even better classification accuracy. This indicates that the spectral attention mechanism can focus well on spectral features and that the neighborhood convolution module can enhance the Transformer's ability to focus on the local spectrum.

TABLE VII
RESULTS OF ABLATION EXPERIMENTS ON IP DATASET, WITH BEST PERFORMANCE IN BOLD

Method	Channel attention	Neighborhood convolution	OA	AA	KAPPA
GTFN	×	×	93.24	96.37	92.29
	√	×	93.69	96.38	92.80
	×	√	93.73	96.43	92.85
	√	√	**94.50**	**96.65**	**93.71**

2) Effect of Network Layers on Experimental Results: In addition to the network structure, the number of layers of the GCN and encoders of Transformer in the model has a crucial impact on the classification performance. The optimal number of layers of the GCN is highly dependent on the sparsity of the constructed adjacency matrix. Too many layers will generally lead to an oversmoothing phenomenon and poor classification results.

Fig. 13 shows that the OA is highest in the three datasets when the GCN uses four, five, and five layers. For the Transformer encoder, the best classification results are achieved when using six, four, and six layers.

3) Effect of Patch Size and Transformer Input Pixels on Classification Accuracy: In addition, this article experimentally investigates the effect of the input patch size and the number of pixels selected by the pixel selection module on the classification results. Fig. 14 shows the effect of different patch sizes on the classification accuracy of the three datasets. Generally speaking, larger patches contain more spatial infor-

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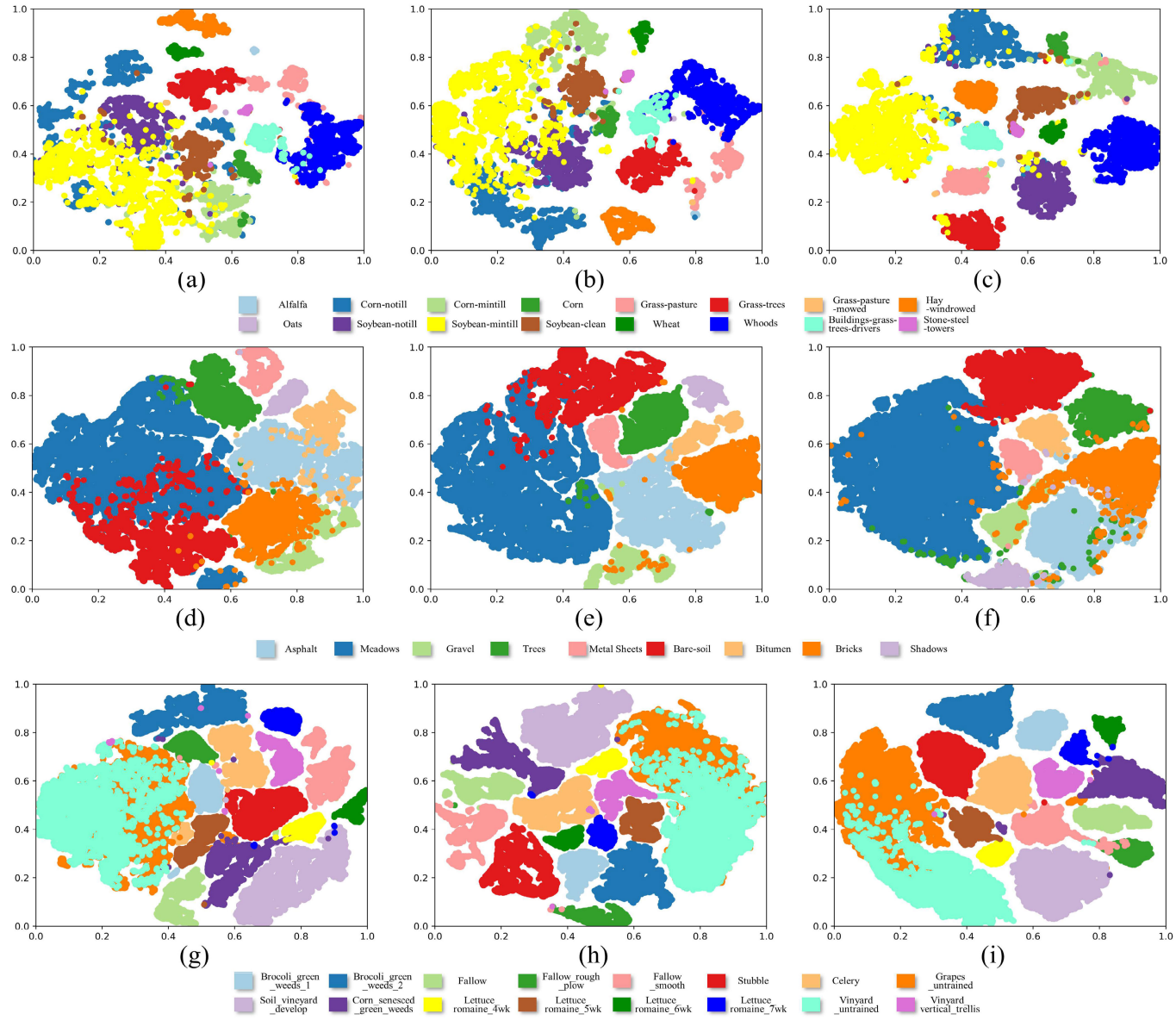


Fig. 11. Feature separability for different methods in the three datasets. (a) SSFTT, (b) GAHT, and (c) GTFN for IP. (d) SSFTT, (e) GAHT, and (f) GTFN for UP. (g) SSFTT, (h) GAHT, and (i) GTFN for SA.

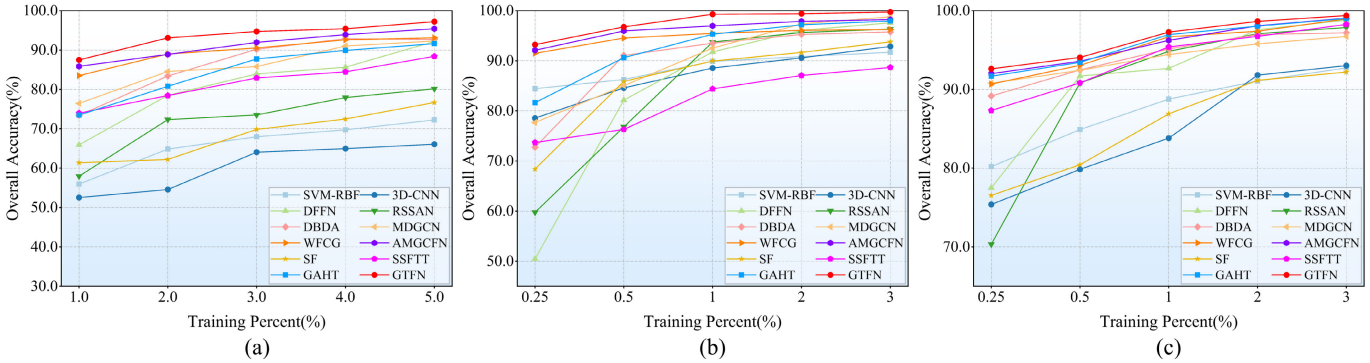


Fig. 12. OA for different proportions of training samples on the three datasets. (a)–(c) OA curves in the IP, SA, and UP datasets, respectively.

mation. On the other hand, larger patches are more likely to introduce more noise, resulting in inaccurate classification and also causing more computational complexity. From the data in the figure, we can see that when the patch size is chosen to be 9×9, it performs even better and achieves the highest classification accuracy in all three datasets. It also shows that

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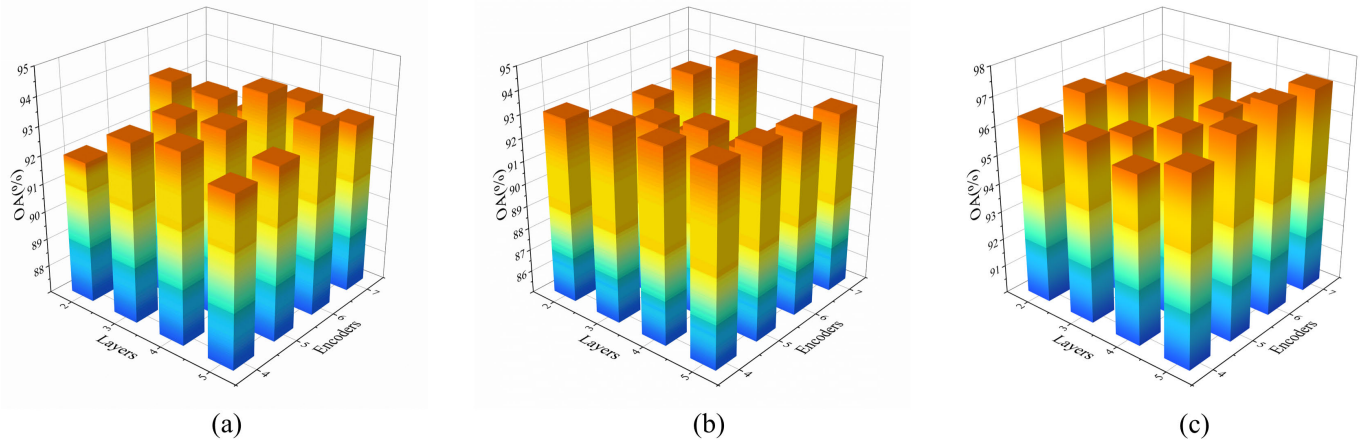


Fig. 13. OA with different numbers of GCN layers and Encoders. (a)–(c) 3-D histograms of OA in the IP, UP, and SA datasets, respectively.

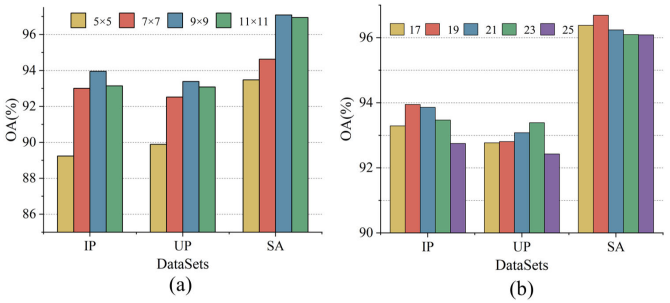


Fig. 14. Left panel shows the OA for three datasets of different Patch sizes. From left to right, the OA bars in the IN, UP, and SA datasets, respectively. The right panel shows the OA of three datasets with different selection pixel sizes. From left to right, the OA bars in the IP, UP, and SA datasets, respectively.

TABLE VIII
COMPARISON OF RUNNING EFFICIENCY FOR DIFFERENT METHODS

Method	Running Time (s)						Complexity (FLOPs)
	IP		SA		UP		
	Train	Test	Train	Test	Train	Test	
3D CNN	76.27	3.16	81.60	16.92	30.27	17.18	0.69G
DFFN	75.68	2.09	78.30	10.99	61.26	21.16	4.27G
RSSAN	57.00	2.08	61.06	9.10	37.34	12.73	0.85G
SF	470.37	46.05	510.01	254.53	255.96	263.03	1.50G
SSFTT	47.87	2.02	52.67	10.43	26.86	10.77	2.59G
GAHT	75.26	2.91	81.18	15.07	48.72	20.26	4.72G
DBDA	640.05	18.48	280.64	35.70	93.10	13.20	1.54G
MDGCN	409.25	1.11	153.10	0.41	747.37	5.18	6.80G
AMGCFN	83.54	4.99	142.24	7.13	199.93	14.15	6.32G
WFCG	204.45	1.61	845.55	7.04	699.76	12.00	5.98G
GTFN	**110.64**	**4.08**	**114.97**	**8.54**	**86.16**	**11.41**	**0.74G**

too large or too small a patch size can affect the classification. In addition, we also analyzed the size of the selected pixels, and the results are shown in Fig. 14; 19, 23, and 19 pixels were selected to achieve the best classification results in the IP, UP, and SA datasets, respectively.

E. Comparison of Computational Efficiency

To compare the operational efficiency and computational consumption of different methods, we tested the training and testing time of the methods in Table VIII on three datasets. We compared the complexity of different models by calculating FLOPs. All methods are tested on the same computing platform. This article designs a joint classification architecture of GCN and Transformer, which improves classification accuracy while increasing the time cost of training and testing. The training and testing time is moderate and within an acceptable range compared to other networks. It is worth noting that this article uses the Follow Patch as the input of the network. Hence, the complexity of the GTFN is relatively tiny, and FLOPs reach a lower level, which also gives GTFN a broader industrial application prospect.

V. CONCLUSION

To fully use HSI's spatial-spectral domain information and realize fine-grained features' classification, we propose a semisupervised classification framework GTFN based on GCN's spatial modeling ability and Transformer's ability to mine spectral domain features. The Follow Patch module is designed to enlarge the receptive field of classified pixels and reduce the computational complexity of GCN. SAGCN is proposed to make GCN pay more attention to critical spectrums more efficiently by introducing the spectral attention mechanism. To solve the problem that transformer feature extraction is scattered, the neighborhood convolution is designed, which improves the ability of the Transformer to pay attention to local features and proposes LST. GTFN can better use spatial and spectral features and has higher classification accuracy than state-of-the-art algorithms. The classification accuracy OAs of GTFN on IP, SA, and UP datasets are 94.00%, 96.81%, and 95.14%, respectively.

GTFN creatively fuses GCN and Transformer, showing its powerful spatial-spectral information extraction capability and broad application prospects. In the future, we will consider exploiting the architecture of this article for unsupervised learning of HSI and further improving the classification performance using more advanced techniques, as well as exploring

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more lightweight networks to reduce the complexity of the network.

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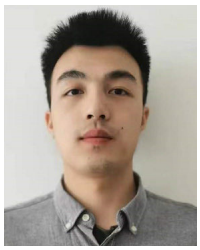
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